

Optical resonance and coherent transients in dressed atomic systems

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We experimentally investigate the transient dynamics of two-level atoms driven by a bichromatic optical excitation field when the bichromatic field is configured to contain a strong resonant component and a weaker nonresonant perturber component. We concentrate on the experimental situation in which the two-level atoms are prepared in an eigenstate (dressed state) of the coupled atom-resonant field system prior to exposure to the perturber field. We experimentally demonstrate perturber-field-induced, optical nutation of the dressed-state populations. We show how nutation of the dressed-state populations manifests itself in the fluorescence signal of the transiently driven atoms. We observe that the detuning of the perturber field from transitions between the dressed states of the atom-resonant field system plays the same role in determining the character of the dressed-state optical nutation as the atom-field detuning in the case of the ordinary optical nutation of atomic populations. Our experiments demonstrate the utility of the dressed-atom picture in understanding the complex dynamics of bichromatically driven atoms in a variety of circumstances.

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I. INTRODUCTION

Coherent interactions between light fields and two-level atoms have been shown to give rise to a number of interesting and useful coherent transient effects such as optical nutation, free-induction decay, photon echoes, and so on [1–5]. In all of these cases, the phenomena have been seen to result from the interaction of a light field with a purely material system (atom, molecule, etc.) [6].

In the present paper, we experimentally investigate the transient excitation of dressed atoms [7–9]. Dressed atoms are composite systems (“molecules”) consisting of a light field and an atom. Their energy levels are the eigenstates of the coupled atom plus field system. We experimentally demonstrate that a secondary excitation field, weak compared to the dressing field, can be employed to excite transitions of the composite light plus atom molecule so as to produce coherent transient effects analogous to those routinely seen in the case of purely material systems.

We note that the dressed-atom concept has a long history. Prior to its application to purely optical systems [7–12] it was employed in various forms to describe systems driven at radio frequencies (rf) [13] and at both optical and rf frequencies [14]. While the standard two-level-atom dressed-atom model is strictly valid only in the context of a single excitation field, in certain limits it provides an excellent basis in which to describe atomic interactions with more than one field. This is demonstrated in the present work and in the recent use of the dressed-atom model to predict and explain two-photon lasing using a driven-atom gain medium [15].

We also note that our experiments can be interpreted directly in terms of bichromatic excitation without introducing the dressed-atom picture. The latter approach has been employed in the context of nuclear magnetic resonance [16]. While the direct analysis is more general,

the driven dressed-atom analysis provides considerable insight into the complex dynamics [17–19] characteristic of bichromatically driven systems.

It is interesting to identify important differences between optical and rf regime studies of driven-atom dynamics. In the rf regime, one is frequently concerned with situations where driving-field Rabi frequencies are comparable to level splittings and the breakdown of the rotating-wave approximation is important. In the optical regime, driving fields of such strength typically lead to ionization. With rf fields, realization of excitation that is homogeneous throughout a sample is fairly simple. In the optical regime, great care must be taken to ensure that millimeter-scale interaction volumes are uniformly excited. In the dressed-atom picture level splitting depends on driving-field strength and a simple level structure thus depends critically on field homogeneity.

II. DEFINITION OF THE SYSTEM

Consider a two-level atom with states $|1\rangle$ and $|2\rangle$ separated in energy by $\hbar\omega_{21}$ [see Fig. 1(a)]. Consider its interaction with a strong, resonant, pump field (the dressing field) and a weaker, secondary field, i.e.,

$$E(t) = (E_p + E_s e^{-i(\Delta t + \Phi)})e^{-i\omega_p t} + \text{c.c.}, \quad (1)$$

where E_p, ω_p (E_s, ω_s) represent the amplitude and frequency of the pump (secondary) field, $\Delta \equiv \omega_s - \omega_p$ is the detuning of the secondary field from the pump field, and Φ is the relative phase difference between the fields at $t=0$. We define Ω_p (Ω_s) as the pump (secondary) field only, bare-atom, resonant Rabi frequency and Γ as the natural width of the bare atomic transition. It is assumed throughout that $\Omega_p \gg \Omega_s$ and that the pump field and atom are exactly resonant, i.e., $\omega_p = \omega_{21}$.

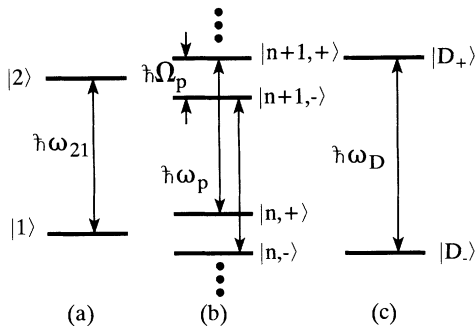


FIG. 1. (a) Two-level atom energy level diagram. (b) Dressed-atom doublets with n representing the number of quanta excited in the atom-pump-field system ($n \gg 1$). (c) A discrete two-level-atom system comprised of pump-field dressed states.

In the dressed-atom approach [7,9], the atom-pump-field interaction is incorporated along with the free-atom Hamiltonian into an expanded Hamiltonian whose eigenstates are the “dressed states.” The energy eigenstates of a dressed atom in a quantized pump field are shown in Fig. 1(b). These states constitute a ladder of doublets separated and split in energy by $\hbar\omega_p$ and $\hbar\Omega_p$, respectively. Provided, as assumed here, that $\Omega_p \gg \Omega_s \gg \Gamma$, one expects the secondary field to interact with distinct transitions of the pump-field dressed atom just as if those transitions were part of a purely material system. In Fig. 1(c), we depict a generic pump-field dressed-atom transition. In our case of experimental interest, the $|D_- \rangle(|D_+ \rangle)$ state corresponds to pump-field dressed states of the type $|n, - \rangle(|n + 1, + \rangle)$, and $\omega_D = \omega_p + \Omega_p$. Note that multiple dressed-atom transitions corresponding to different values of n are excited simultaneously.

III. EXPERIMENT

A schematic diagram of our apparatus is shown in Fig. 2(a). A cw ring dye laser (RDL) is frequency locked to a magnetically tunable Lamb dip of a Yb reference cell. The main laser output (about 750 mw) is passed through two acousto-optic modulators (AOM's) each driven by a bichromatic rf field. The bichromatic rf field is produced by combining the output of two separate and free-running rf oscillators, one of which is frequency tunable. Details of the interactions between the optical beam(s) and the bichromatic rf field inside the two AOM's are given elsewhere [19]. The optical field emergent from the sequence of two AOM's is spectrally analyzed by a 50-cm confocal Fabry-Pérot interferometer. Although the optical excitation field created is predominately bichromatic with one strong and one weak frequency component acting as the pump and secondary driving field, respectively, it possesses a small third frequency component which is four times lower in power than the secondary field. Calculations show that this third component has neg-

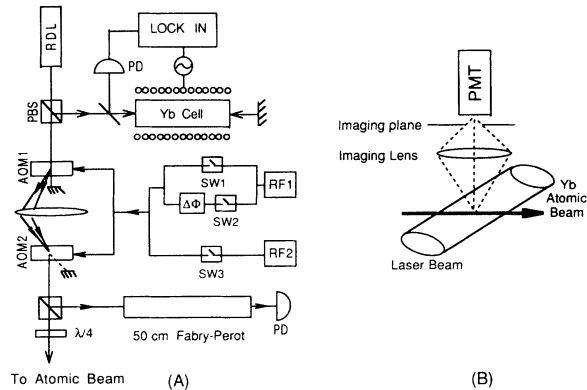


FIG. 2. (a) Experimental apparatus for pump and secondary optical field generation. RDL, ring dye laser; PBS, polarization beam splitter; AOM, acousto-optic modulator; SW, rf switch; $\Delta\phi$, rf phase shifter. The Fabry-Pérot interferometer spectrally analyzes the final optical field for diagnostic purposes. (b) Schematic of the atom-field interaction and the signal collection apparatus. PMT, photomultiplier tube.

ligible impact on atomic behavior under the conditions addressed by our experiment. Since both the pump and secondary laser fields originate from the same laser source and travel along essentially the same optical path, their relative phase is fixed by the relative phase of the bichromatic rf field components used to create them. The importance of the relative phase of the fields arises from the fact that for certain initial conditions (i.e., when the $|D_- \rangle$ and $|D_+ \rangle$ states are prepared in a coherent superposition), it can influence the transient dynamics of the driven dressed atom [17,18]. In the present experiment, dressed atoms are prepared so as to reside entirely in one terminal level of the dressed-state transition excited, making the dynamics essentially insensitive to the initial relative phase of the excitation components.

Transient amplitude- and phase-controlled excitation fields (10-nsec transition times) are created [see Fig. 2(a)] by gating the AOM drive fields with rf switches SW1, SW2, and SW3. Abrupt changes in the pump-field phase, necessary to place the system in a single dressed level, are introduced by appropriate switching of SW1 and SW2. The magnitude of the phase shift introduced in the pump field is controlled by a rf phase shifter. Attenuators used to control the amplitude of the rf (and hence optical) fields are not shown.

The experiments are performed in a beam of atomic ^{174}Yb utilizing the $^1S_0 - ^3P_1$ atomic transition at 556 nm. The atoms are nuclear-spin-free so they behave as nearly ideal two-level atoms when excited by circularly polarized light. The radiative lifetime of the excited 3P_1 state is 875 nsec. To minimize the Doppler broadening, the atomic beam is well collimated and intersects the laser beam at right angles [Fig. 2(b)]. Scans of the ^{174}Yb absorption profile reveal a 3-MHz residual Doppler linewidth. To ensure uniform laser intensity throughout the ^{174}Yb -laser interaction region, a large laser beam (3.0 mm) and a small atomic beam (0.25 mm) are employed. An

imaging system limits signal collection to the intensity-uniform, central region of the laser beam. The atomic fluorescence photons are detected by a cooled photomultiplier tube (PMT).

Our experimental observations consist of measurements of the time-resolved atomic fluorescence (which is proportional to the bare-atom excited-state population, n_e) during transient excitation sequences. In each experimental cycle, the initially ground-state atoms are first exposed to the resonant pump field ($\Omega_p/2\pi = 15$ MHz) until an excitation area of $\pi/2$ accrues. The phase of the pump field is then abruptly shifted by 90° . Subsequent to the pump-field phase shift, the dressed atoms reside in the $|D_-\rangle$ dressed level [20] [see Fig. 1(c)]. The secondary field ($\Omega_s/2\pi \cong 1.5$ MHz) is turned on immediately after the pump-field phase shift with $\Delta/2\pi \approx 15$ MHz so that it is resonant or slightly detuned from the transition between the $|n, -\rangle$ ($|D_-\rangle$) and $|n+1, +\rangle$ ($|D_+\rangle$) dressed states. Fluorescence measurements, made throughout the on time of the secondary excitation field, span approximately two lifetimes of the bare-atom excited state. The excitation sequence is repeated at approximately 200 KHz. This rate is limited by the approximately 1.5- μ sec duration of the excitation fields and the time necessary for the sample to relax back to the bare-atom ground state. The average number of fluorescence photons recorded in a particular cycle was significantly smaller than unity. Accordingly, the temporal evolution of the fluorescence was acquired by recording detected photon arrival times over a large number of experimental cycles. Acquisition of the complete temporal evolution of the fluorescence for a single secondary field detuning required about 90 sec.

IV. EXPECTED DYNAMICS

Prior to presenting our experimental results, we utilize our driven dressed-atom model to gain insight into the types of behavior that the bichromatically driven Yb atoms might be expected to display. The dynamics of the system are simplest when viewed in terms of dressed-atom quantities, e.g., dressed-state population. Unfortunately, these quantities were not directly observable in our experiments, and information about the driven dressed-atom dynamics must therefore be inferred from the experimentally accessible information.

Consider an atom exposed to the resonant pump field and in a linear superposition of dressed states. Assume that the n -summed, upper and lower dressed-state populations are Π_+ and Π_- , respectively. Neglecting relaxation and in the presence of the pump field only, the atom's dressed-state superposition (and hence Π_+ and Π_-) will be time independent. However, using the well known relationships between the dressed and bare atomic states [7], one can show that a time independent linear superposition of dressed states leads to time-dependent bare-state behavior. The atom will exhibit, for example, an excited-state population oscillation (our primary experimental observable) given by

$$n_e(t) = \frac{1}{2}[1 + (1 - W_D^2)^{1/2} \cos(\Omega_p t + \phi)], \quad (2)$$

where $W_D \equiv \Pi_+ - \Pi_-$ is the inversion in the dressed-atom basis. In the absence of the secondary field and neglecting relaxation, W_D is constant. We see that the bare-atom, excited-state population will oscillate at the pump-field Rabi frequency. This oscillation reflects optical nutation of the bare atomic populations and is familiar from studies of transient single-frequency optical excitation [6]. The oscillation's phase ϕ depends in detail on the initial state of the system. Importantly, as seen in Eq. (2), the amplitude of the excited-state population oscillation is dependent upon the dressed-state inversion. When an atom is in a pure dressed state (i.e., $W_D = \pm 1$), bare-atom optical nutation is completely suppressed [20,21].

In the presence of the secondary driving field, the situation is richer and more complex. In the limits of interest here ($\Omega_p \gg \Omega_s \gg \Gamma$ and $\Omega_p \gg \delta$, where $\delta = \omega_s - \omega_D$), the secondary field can be considered to act in isolation on a single transition of the pump-dressed atom. Accordingly, dressed-state observables should respond to step turn-on secondary-field excitation just as analogous bare-atom quantities do in the presence of step turn-on monochromatic excitation. With this fact in mind and assuming that the pump-dressed atoms have an initial inversion of $W_D = -1$, one can immediately write

$$W_D(t) = -\left(\frac{\Omega_D}{\Omega'_D}\right)^2 \cos(\Omega'_D t) - \frac{1}{2} \left[1 + \frac{(\delta^2 - \Omega_D^2)}{\Omega_D'^2}\right], \quad (3)$$

where $\Omega_D'^2 = \Omega_D^2 + \delta^2$ and, for a resonant pump field, $\Omega_D = \Omega_s/2$. We emphasize that Eq. (3) is identical in form to the expression for bare-atom inversion obtained when a ground-state atom is subjected to step monochromatic excitation. Interestingly, however, the effective Rabi frequency Ω_D of the secondary field acting resonantly on the dressed-atom transition is only half of what it is when the secondary field acts on the bare-atom transition. This fact can be motivated by noting that the electric-dipole matrix elements of the dressed- and bare-atom transitions are in the ratio 0.5:1. When the secondary field is detuned from the dressed-atom transition, i.e., $\delta \neq 0$, the dressed-atom optical nutation rate occurs at the generalized dressed-atom Rabi frequency Ω'_D .

If the secondary field causes W_D to evolve temporally as indicated in Eq. (3), then Eq. (2) indicates that bare-atom excited-state fluorescence should exhibit a high frequency oscillation at Ω_p that is modulated at the slower dressed-state optical nutation frequency Ω'_D . This behavior is examined in more detail in Fig. 3, where $n_e(t)$, calculated using Eqs. (2) and (3), is plotted for various parameters characteristic of our experimental system. An interesting feature is observed as the detuning δ is increased relative to Ω_D . One sees that alternate minima in the $n_e(t)$ envelope disappear. An analogous behavior could be observed in the optical nutation of monochromatically driven bare atoms. To see it, however, one must

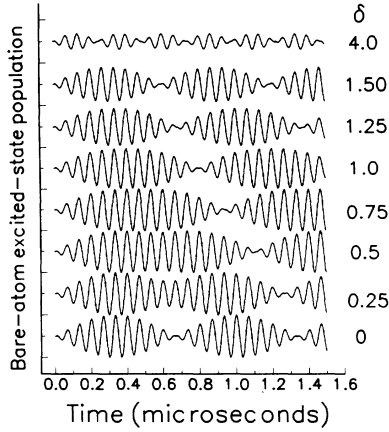


FIG. 3. Representative temporal evolution of excited bare-state atomic population calculated using Eqs. (2) and (3). Parameters employed are $\Omega_p/2\pi = 15$ MHz, $\Omega_s/2\pi = 1.5$ MHz, and those δ values shown. Excited-state population varies from 0 to 1 as the traces span an interval between large vertical tick marks.

observe the time-dependent amplitude of the bare-atomic dipole moment rather than the bare-atom excited-state population.

We have found that the low frequency modulation of the $n_e(t)$ oscillation follows from the Rabi flopping of population back and forth between the dressed states. For large δ , the envelope function exhibits a node every time the dressed atom has returned to its initial state. As in the case of off resonant monochromatic excitation, the $\delta \neq 0$ secondary field never drives the dressed atoms entirely out of the initially occupied state. For small δ , the dressed atoms actually evolve through the initially unpopulated dressed state leading to the appearance of additional nodes in the $n_e(t)$ oscillation.

V. OBSERVATIONS AND DISCUSSION

In Figs. 4(a)–4(d), we show experimental measurements of $I_f(t)$, the atomic fluorescence intensity versus time [proportional to $n_e(t)$], for various values of δ . As expected on the basis of Eqs. (2) and (3), $I_f(t)$ displays, for all δ , an approximately 15-MHz oscillation with a slowly damped oscillatory envelope. The overall damping results from experimental imperfections such as residual laser-field inhomogeneity, atomic spontaneous emission, residual Doppler broadening, and transient time. Laser-field inhomogeneity is particularly important since it leads to spatial variations in Ω_p and concomitant dephasing of the nutating atoms.

In Figs. 4(e)–4(h), calculations of $n_e(t)$ performed using Eqs. (2) and (3) and the experimental parameters of Figs. 4(a)–4(d) are presented. Qualitative agreement between the predicted and observed behavior is clear. Detunings small enough to probe the disappearance of alternate envelope minimum (see Fig. 3) were not stud-

ied. Nevertheless, we do see agreement of predicted and calculated behavior for both the resonant ($\delta = 0$) case (which has extra envelope minima) and the nonresonant cases (which, since $|\delta| > \Omega_D$, do not have extra minima).

It is interesting to examine the reliability of the predictions obtained from our simple dressed-atom model of bichromatic excitation. To do this, we have calculated the atomic dynamics expected under our bichromatic excitation conditions by direct numerical integration of the bare-atom equations of motion. These numerical simulations of $n_e(t)$ are shown in Figs. 4(i)–4(l). One immediately notes that the behavior predicted by direct numerical integration of the bare-atom equations of motion is essentially indistinguishable from that predicted analytically by modeling the problem in terms of monochromatic excitation of an isolated dressed-atom transition. It is reasonable to assume that the single-transition, dressed-atom model will work well whenever the dressed-atom transitions are well separated compared to their widths, the secondary-field Rabi frequency Ω_D , and the detuning of the secondary field from a particular dressed-atom transition, δ . The same model should be useful for cases of bichromatic excitation in which the pump field is not resonant with the bare-atom transition. In such cases,

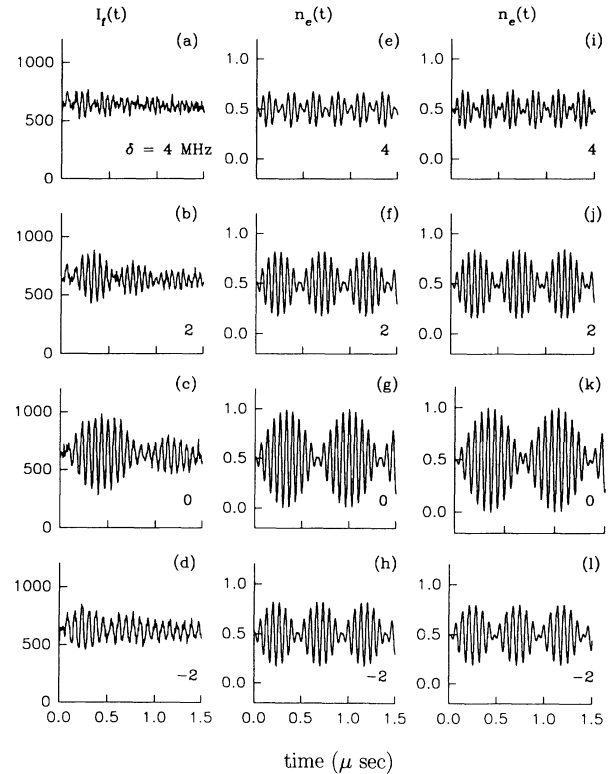


FIG. 4. (a)–(d) Recordings of atomic fluorescence intensity versus time, $I_f(t)$. The detuning of the secondary field from the dressed-state transition frequency δ is indicated in the lower right corner of each trace. (e)–(h) Excited-state atomic population versus time, $n_e(t)$, calculated using Eqs. (2) and (3) ignoring all damping mechanisms. (i)–(l) $n_e(t)$ calculated by numerical integration of the bare-atom equations of motion.

Ω_D will vary depending on the dressed-atom transition and will in general differ from $\Omega_s/2$. In contrast to the case of a resonant pump, the steady-state, dressed-state populations for a nonresonant pump are polarized. A secondary field resonant with a particular dressed-atom transition can equalize the terminal-level populations, and therefore perturb the normal steady state. This fact can be utilized, for example, in optical trapping, cooling, and/or manipulation since the dipole force experienced by an atom is related to the distribution of population among the dressed levels.

For certain detunings of the secondary field from the pump field, it is possible to excite multiphoton dressed-state transitions. In such cases, the dressed atom is driven between components of dressed-state doublets differing in n [see Fig. 1(b)] by more than one [15,22–26]. Simple, isolated transition, dressed-atom models should also permit insight into the dynamics created by transient excitation of multiphoton dressed-atom transitions.

VI. SUMMARY

To summarize, we have experimentally explored the transient dynamics of two-level atoms driven by bichromatic excitation. We have considered parameter ranges wherein the dynamics observed can be simply described in terms of the dynamics of monochromatically driven dressed atoms. Owing to the apparent complexity of bare-atom dynamics in the presence of bichromatic excitation, the dressed-atom perspective is especially significant. Steady-state extensions of the dressed-atom approach to treating bichromatic excitation may prove useful in diverse fields such as optical atom cooling and trapping [27].

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